

$T_0 = y_t$
 $CS = y_t^c$
 $P = y_t + t$

Sales.

1	20
2	16
3	12
4	8
5	4
6	0
7	-4

Forecast

Approaches $\begin{cases} \text{Qual} \\ \text{more than binary} \end{cases}$

Types $\begin{cases} LT \rightarrow 3 - \text{down} \\ ST \rightarrow 3 - \text{down} \\ MT \rightarrow 3 - 6 \end{cases}$

Uses.

Benefits $\begin{cases} \rightarrow \text{descriptive (Tup)} \\ \rightarrow \text{spatial (C.S)} \\ \rightarrow \text{explorative} \end{cases}$

relationship b/w two

Application:

1) Economic
 2) Sales
 3) Performance
 4) Portfolio
 5) Demand
 6) Yield.

20/08/26

U-T

- TS
- Graph
- Add by multiplication

* components (decomposition)

* Data transformation

* trend $\begin{cases} \text{linear} \\ \text{quadratic} \\ \text{Exponential} \end{cases}$

Advantages (linear)

- 1) Simple
- 2) Short run

Disadvantages

- 1) Assumes constant slope
- 2) +ve data

$y = a + bt + ct^2 \rightarrow 2^{\circ} \text{ polynomial}$

$= a + bt + ct^2 + dt^3 \rightarrow 3^{\circ} \text{ polynomial}$

Extrapolation $\rightarrow$ outside the data

Exponential

Adv:

- * Financial analysis
- * compound v. growth rate

Dis

- * ex
- * +ve.

y	Q1	Q2	Q3	Q4
21	3.7	4.7	3.3	3.5
22	3.7	3.9	3.6	3.6
23	4.0	4.1	3.3	3.1
24	3.3	4.4	4.0	4.0

3.675 4.275 3.55 3.55 $\rightarrow$ (Avg) =

Seasonal $\frac{g(A1)}{q_A} \times 100$ $\frac{g(A2)}{q_A} \times 100$ $\frac{g(A3)}{q_A} \times 100$ $\frac{g(A3)}{q_A} \times 100$

0.976 1.186 0.943 0.943

97.6	118.26	94.3	94.3
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y_t	y_{t-1}	y_{t-2}	$(y_t - y_{t-1})$	$(y_{t-1} - y_{t-2})$	$\Delta^2 y_t$
10	-	-	10	-	10
12	10	-	2	10	-8
15	12	10	3	2	1
19	15	12	4	3	1
25	19	15	6	4	2
28	25	19	3	6	-3
32	28	25	4	3	1

Growth rates

Absolute growth rate = $y_t - y_{t-1}$

Relative growth rate = $\frac{y_t - y_{t-1}}{y_{t-1}}$

% growth rate = $\left(\frac{y_t - y_{t-1}}{y_{t-1}} \right) \times 100$

24/08/26

	Q1	Q2	Q3	Q4	
2022	4.8	4.1	6.0	6.5	
2023	5.8	5.2	6.8	7.4	
2024	6.0	5.5	7.5	7.8	
2025	6.3	5.9	8.0	8.4	overall mean
mean	5.7	5.2	7.1	7.4	6.4
Seasonal	0.897	0.815	1.109	1.179	

$y_t - y_{t-1} / y_{t-1} = \frac{\Delta y_t}{y_{t-1}} = \frac{\Delta y_t}{y_{t-1}}$

Abs. G.R.

y_t	y_{t-1}	y_{t-2}	$\Delta y_t (y_t - y_{t-1})$	Δy_{t-1}	$\Delta^2 y_t$	Rel. G.R.
7	0	0	-	0	-	-
12	7	0	5	7	-2	0.7143
15	12	7	3	5	-2	0.2500
19	15	12	4	3	1	0.2667
25	19	15	6	4	2	0.3158
32	25	19	7	6	1	0.2800
45	32	25	13	7	6	0.4063
49	45	32	4	13	-9	0.0889
54	49	45	5	4	1	0.1020

01/09/26

* Easy

* Reduces the effects of extreme pt

* Makes trend visible

* Reduces Random Fluct

Smoothing method $\begin{cases} \text{MA} \\ \text{Ex. Smo. method} \end{cases}$

Exp. SM $\begin{cases} \text{simple} \\ \text{double} \\ \text{triple} \end{cases}$

$$F_{t+1} = \cancel{F_t} + \alpha (A_t - F_t)$$

$\xrightarrow{\text{forecasting value}}$ $\xrightarrow{\text{current value}}$

$$\alpha = 20 + 0.05(20 - 20)$$

$$= 27 + 0.05(\alpha - 27)$$

$$F_2 = F_1 + \alpha(A_1 - F_1)$$

$$F_3 = F_2 + \alpha(A_2 - F_2)$$

$$F_4 = F_3 + \alpha(A_3 - F_3)$$

α is used when there is no trend in sample

$$F_{t+1} = \cancel{y_t} + \alpha(A_t - F_t)$$

Sales.

$$20 \quad F_2 = 20 + 0.05(20 - 20) = 20$$

$$27 \quad F_3 = 20 + 0.05(27 - 20) = 20.35$$

$$25 \quad F_4 = 20.35 + 0.05(25 - 20.35) = 20.58$$

$$22 \quad F_5 = 20.58 + 0.05(22 - 20.58) = 20.65$$

$$18 \quad F_6 = 20.65 + 0.05(18 - 20.65) = 20.52$$

23

27

~~23~~

$$F_2 = F_1 + 0.05(A_1 - F_1)$$

$$F_2 = 20 + 0.05(20 - 20)$$

$$F_3 = F_2 + 0.05(A_2 - F_2); 20 + 0.05(27 - 20)$$

02/09/20

Day	1	2	3	4
M	43	51	40	64
T	45	41	57	58
W	22	37	30	33
T	25	22	33	38
F	31	25	37	25

- compute 3 days moving average
- plot the series on the moving average on the graph
- Does it appear to be seasonal (weekly pattern)?

21

26

19

16

28

25

26

27

39 37 61 58 18 52 82 27 41 69

49 66 54 42

$\alpha = 0.2$

Simple Exponential Smoothing method with $\alpha = 0.2$.

03/09/26

$F_{t+1} \rightarrow$ next period forecast

$F_t \rightarrow$ current year forecast

$Y_t \rightarrow$ Actual Value

α - Smoothing constant $0 < \alpha < 1$

$$F_{t+1} = F_t + \alpha(A_t - F_t)$$

$$(or) F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

$$F_{t+1} = \alpha A_t + (1-\alpha)F_t$$

$\alpha = 0.8$ Forecast reacts quickly to changes (Rapid Response)

$0.2 \rightarrow$ slow & smooth.

$$g_0 = \alpha = 0.5$$

$$\text{Jun actual} = A_{\text{Jun}} = 130$$

$$F_{\text{Jun}} = 115$$

Calculate: F_{July}

$$F_{\text{July}} = F_{\text{June}} + 0.5[A_{\text{Jun}} - F_{\text{Jun}}]$$

$$F_{\text{July}} = 115 + 0.5[130 - 115]$$

$$= 122.5$$

$$\alpha = 0.2 \text{ at } F_{\text{Jan}} = 100$$

Jan 100

$$F_{\text{Feb}} = 100 + 0.2(100 - 100) = 100$$

Feb 120

$$F_{\text{Mar}} = 100 + 0.2(120 - 100) = 104$$

Mar 130

$$F_{\text{Apr}} = 104 + 0.2(130 - 104) = 109.2$$

Apr 125

$$F_{\text{May}} = 109.2 + 0.2(125 - 109.2) = 112.36$$

May 140

$$F_{\text{June}} = 112.36 + 0.2(140 - 112.36) = 117.87$$

June ?

2021 Q1 4.8

2022 Q1 5.8

2023 Q1 6

Q2 4.6

Q2 5.2

Q2 5.6

Q3 6

Q3 6.8

Q3 7.5

Q4 6.5

Q4 7.4

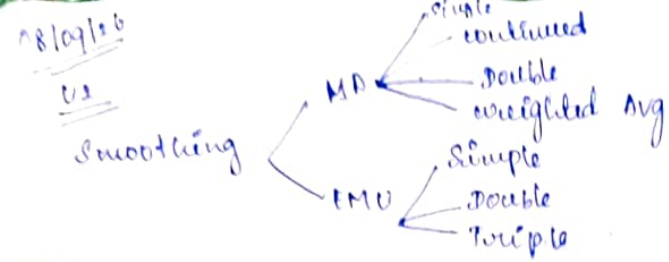
Q4 7.8

2024 Q1 6.3

Q2 8

Q2 5.9

Q4 8.4



IMU

Simple: $F_{t+1} = F_t + \alpha (A_t - F_t)$

Double $\rightarrow$ $S_t = \alpha A_t + (1-\alpha)(S_{t-1} + T_{t-1})$ $0 \leq \alpha \leq 1$

(Holt method) $T_t = \beta (S_t - A_{t-1}) + (1-\beta)T_{t-1}$ $0 \leq \beta \leq 1$

$F_{t+1} = S_t + T_t$

Triple

HOLT WINTERS METHOD

A) H-w additive method

$L_t = \alpha (y_t - S_{t-2}) + (1-\alpha)(L_{t-1} + b_{t-1})$

$b_t = \beta (L_t - L_{t-1}) + (1-\beta)b_{t-1}$

$S_t = \gamma (y_t - L_t) + (1-\gamma)S_{t-1}$

$F_{t+m} = L_t + mb_t + S_{t-2+m}$

B) H-w multiplicative:

Levels $L_t = \alpha \frac{y_t}{S_{t-2}} + (1-\alpha)(L_{t-1} + b_{t-1})$

Trend $b_t = \beta \left(\frac{L_t}{L_{t-1}} \right) + (1-\beta)b_{t-1}$

$S_t = \gamma \left(\frac{y_t}{L_t} \right) + (1-\gamma)S_{t-1}$

$F_{t+m} = (L_t + mb_t) S_{t-2+m}$

Holt winter's Exponential smoothing techniques has three components

1) Levels $\rightarrow$ It gives the curr. value

(Eq) curr. Sales = 500

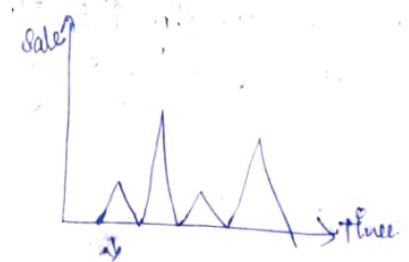
2) Trends $\Rightarrow$ ~~Sales~~ the sales going up or coming down?

(Eq) Sales $\uparrow$ by 20 units every month.

3) Seasonality $\Rightarrow$ Repeated pattern

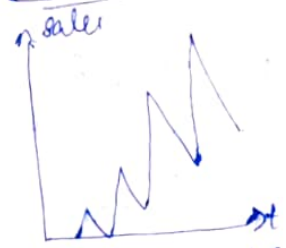
(Eq) Dec sales are usually higher than others in every year.

Additive sales



Seasonal variations are same in additive model

multiplicative model



Seasonal variation becomes larger over time as the time goes.

Comparison of additive or multiplicative model of Holt winter's technique.

Feature	A	M
Seasonal effect	constant	It changes with levels
Formula	Level + Seasonality	Level $\times$ Seasonality

Problem:-
cat.
multiplicative Holt winter's technique

Suppose a hotel has the current level to be 1000 booking. trend + 100. Dec seasonal effect is +200. forecast

Soln

$$L+T+S = 1000 + 100 + 200 = 1300$$

2) Assume Dec month usually all the booking by 20% so the forecasting will be

$$\begin{aligned} \text{Forecast} &= (L+T) \cdot S \\ &= (1000 + 100) \cdot 0.2 \\ &= 220 \end{aligned}$$

Note:- $\rightarrow$ Additive.
 Holt winter's uses seasonal variation to be roughly constant in magnitude i.e. add the same amt

In multiplicative, the seasonal variation changes proportionally with the levels of a time series data i.e. multiplied by some percentage

Given monthly or quarterly data calculate the seasonality of the seasonality at

In multiplicative model Seasonal index is a %10.

10/09/26 ① De-seasonal

	Q1	Q2	Q3	Q4
2011	3.7	4.7	3.3	3.5
12	3.7	3.9	3.6	3.6
13	4.0	4.1	3.3	3.1
14	3.3	4.4	4.0	4.0
	<u>3.675</u>	<u>4.275</u>	<u>3.55</u>	<u>3.55</u>
= 3.7625				

SI Q1 =	$\frac{Q1 \text{ Avg}}{\text{Total Avg}}$	SI Q2 =	$\frac{Q2 \text{ Avg}}{\text{Total Avg}}$	SI Q3 =	$\frac{Q3 \text{ Avg}}{\text{Total Avg}}$	SI Q4 =	$\frac{Q4 \text{ Avg}}{\text{Total Avg}}$
	0.977		1.136		0.944		0.944

Sales SP Discount(a/b)

Q1	3.7	0.976	3.790
Q2	4.7	1.13	4.159
Q3	3.3	0.94	3.511
Q4	3.5	0.94	3.723
Q1	3.7	0.976	3.791
Q2	3.9	1.13	3.451
Q3	3.6	0.94	3.829
Q4	3.6	0.94	3.829
Q1	4.0	0.976	4.098
Q2	4.1	1.13	3.628
Q3	3.8	0.94	3.510
Q4	3.1	0.94	3.297
Q1	3.3	0.976	3.381
Q2	4.4	1.13	3.893
Q3	4.0	0.94	4.255
Q4	4.0	0.94	4.255

Exponential Smoothing method (single)

$F_{t+1} = F_t + \alpha(A_t - F_t) \quad \alpha = 0.5$

A_t	F_t	F_{t+1}	α
1.2	1.2	1.2	0.5
0.9	1.2	1.2	0.5
1	1.05	1.05	0.5
0.7	1.025	1.025	0.5
1.3	0.86	0.86	0.5

Forecast 1.08 1.3

Double

$$S_t = \alpha A_t + (1-\alpha)(S_{t-1} + T_{t-1})$$

$$T_t = \beta(S_t - S_{t-1}) + (1-\beta)T_{t-1}$$

$$F_t = S_t + T_t \quad \alpha = 0.5 \quad \beta = 0.5$$

Double	S	T	F
64	64	0	64
82	73	4.5	
92	87.5		
95			
106			

11109126 SP Discount(a/b)

Q1 4.8	Q1	Q2	Q3	Q4
Q2 4.1				
Q3 6	4.8	4.1	6	6.5
Q4 6.5				
Q1 5.8	5.8	5.2	6.8	7.4
Q2 5.2				
Q3 6.8	6	5.6	7.5	7.8
Q4 7.4				
Q1 6	5.53	4.96	6.76	7.23
Q2 5.6				
Q3 7.5				
Q4 7.8				
6.125				

$$\sum_{n=1}^{\infty} \frac{1}{n} = 6.12$$

SP = 0.903 Q2 = 0.811 Q3 = 1.104 Q4 = 1.180

			SP	DSI
Q1	4.8	5.53	0.903	5.31
Q2	4.1	4.96	0.811	5.05
Q3	6	6.76	1.104	5.43
Q4	6.5	7.23	1.180	5.80
Q1	5.8	5.53	0.903	6.42
Q2	5.2	4.96	0.811	6.41
Q3	6.8	6.76	1.104	6.159
Q4	7.4	7.23	1.180	6.27
Q1	6	5.53	0.903	6.64
Q2	5.6	4.96	0.811	6.85
Q3	7.5		1.104	6.79
Q4	7.8		1.180	6.610

	MA	MA(3)	DMA	EMA
MA	8.7			
	4.7			
2.7	3.3			
	3.5			
4.7	3.7			
3.3	3.9			
8.5	3.6			
3.7	3.6			
	4.0			
3.9	4.1			
3.6	3.3			
3.6	3.1			
4.0	3.3			
4.1	4.4			
3.3	4.0			
3.1				
3.3				

15109126

⑦ Sales x $y = -255x + 222900$

22600	1	Forecast using trend data
22400	2	
22250	3	
21800	4	

⑧ y t

8.2 1 $y = 3.819e^{889t}$

9.4 2 Linear Exponential Quadratic

11.8 3

14.2 4

How to choose a best model?

Based on the error only

⑨ Exponential smoothing method (single)

10 $F_{t+1} = F_t + \alpha (A_t - F_t)$

8 $\alpha = 0.5$

9

10

FSM F+1

At	100
10	10
8	10
9	9
11	9
10	10

$$F_{t+1} = \bar{I}_0$$

$$F_{\text{June}} = F_{\text{May}} + 0.5(A_{\text{May}} - F_{\text{May}})$$

Given $\hat{A}_{\text{May}} = 100$

$F_{\text{May}} = 130$

$F_{\text{June}} = 115$

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{\hat{y}_t} \right|$$

$$\text{MAD} = \frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)$$

$$\text{MSD} = \frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2$$

MAPE → Mean Absolute Percentage Error
 MAD → Mean Absolute Deviation
 MSD → Mean Squared Deviation.

16/09/20

HW method is used for forecasting data that contains level, trend or ~~season~~ seasonal pattern to predict future values.

HW → current sales + trend (T(t)) + seasonal effect/pattern.

Retail → Hotel → Education → Airline → Electricity → Hospitals → Banking → E-commerce.

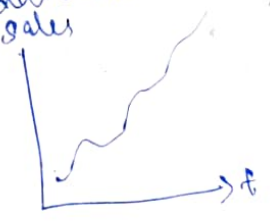
Additive → Seasonal effect remains the same constant [100 units every year + 100]

[+8]
 multiplicative - seasonal effect changes (+10%) with overall levels (*8)

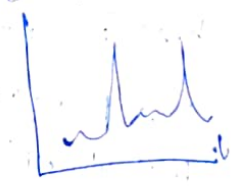
If actual sales is 1000, sales in dec 1000 (20% high than normal)

graphs:

Additive sales



multiplicative sales



Result

curr situations (curr value of data)
curr sales is 500 if the dataset is 100-200-300-400-500

2: controls how $(\frac{\sigma}{\mu})$ the value given for recent observation

→ Trend

Data $\uparrow \downarrow$ [Dirn. of movement matters]
 β : how quickly the estimate trend changes

→ seasonality - repeated pattern over time
 γ : how quickly the seasonal pattern changes

Additive:

$$I_t = \alpha(y_t - S_{t-m}) + (1-\alpha)(I_{t-1} + T_{t-1})$$

$$T_t = \beta(I_t - I_{t-1}) + (1-\beta)T_{t-1}$$

$$S_t = \gamma(y_t - I_t) + (1-\gamma)S_{t-m}$$

$$F_{t+h} = I_t + hT_t + S_{t-m+h}$$

h → Forecasting period

m → seasons in a year.

t → time.

α, β, γ → smoothing coefficient
 $0 \leq \alpha, \beta, \gamma \leq 1$

Additive Seasonal Index

$$IS = \text{Avg } Q_1 - T_A$$

	Q_1	Q_2	Q_3	Q_4
2024	100	120	140	160
2025	120	140	160	180

$$T_A = \frac{1120}{8} = 140$$

Multiplicative:

$$Q_1 = \frac{110}{140} =$$

$$Q_2 = \frac{130}{140} =$$

$$Q_3 = \frac{150}{140}$$

$$Q_4 = \frac{170}{140}$$

Additive:-

$$Q_1 = -30$$

$$Q_2 = -10$$

$$T = ?$$

$$L_8 = 180$$

$$S_8 = 30$$

$$Q_3 = 10$$

$$Q_4 = 30$$

$$L = ?$$

$$L_9 = L_8 (\pm S)$$

$$T_9 = L_9 - (\text{overall avg})$$

$$L_8 - F_8 = 180$$

$$F_9 = L_8 + S_8 + T_8$$

W.W. 1st log h

current level = 100

Trend = 5

Per Quarter Season = 0.94 (M)

Forecast HW = (100+5) * 0.9 → multiplicative

MA
ESM

ADDITIVE - Seasonal effect is added
MULTIPLICATIVE - Seasonal effect is multiplied

ADD:-

$$L_t = \alpha (y_t - s_{t-m}) + (1-\alpha) (L_{t-1} + T_{t-1})$$

$$T_t = \beta (L_t - L_{t-1}) + (1-\beta) T_{t-1}$$

$$s_t = \gamma (y_t - L_t)$$

$$s_t = \frac{Q-m}{T_0}$$

$$F_{t+h} = L_t + hT_t + s_{t-m+h}$$

$$s_t = \frac{Q_1}{T_0}$$

MULT:-

$$L_t = \alpha \left(\frac{y_t}{s_{t-m}} \right) + (1-\alpha) (L_{t-1} + T_{t-1})$$

$$T_t = \beta (L_t - L_{t-1}) + (1-\beta) T_{t-1}$$

$$s_t = \gamma \left(\frac{y_t}{L_t} \right) + (1-\gamma) s_{t-m+h}$$

	Q1	Q2	Q3	Q4
Y1	100	120	110	130
Y2	110	130	120	140
SI _m	0.87	1.04	0.95	1.12

SI_m

-15

5

-5

15

$$Q_1 = \frac{100+110}{2} = 105$$

$$SI_1 = 105 \div 120 = 0.87$$

$$Q_2 = \frac{120+130}{2} = 125$$

$$SI_2 = 125 \div 120 = 1.04$$

$$Q_3 = \frac{110+120}{2} = 115$$

$$SI_3 = 115 \div 120 = 0.95$$

$$Q_4 = \frac{130+140}{2} = 135$$

$$SI_4 = 135 \div 120 = 1.125$$

$$SI_1 = 105 - 120 = -15$$

$$SI_2 = 125 - 120 = 5$$

$$SI_3 = 115 - 120 = -5$$

$$SI_4 = 135 - 120 = 15$$

L_t → levels

y_t → Actual observation

T_t → Trend

m → no. of season

s_t → Seasonal component

W.W.

current level = 100

If Decimals multi

If wh. no → Add

Trend = 5

If I have to calculate 2nd quarter

SI = 5

$$100 + (2 \times 5) + 5 \Rightarrow 115$$

?

	Q1	Q2	Q3	Q4	
Y1	68	62	61	63	SI _m = ?
Y2	65	58	66	61	
Y3	68	63	63	67	SI _m = ?

	A	Δ	SP
2023	Q1	100	0.875
	Q2	120	1.04
	Q3	110	0.958
	Q4	130	1.125
2024	Q1	110	0.875
	Q2	130	1.04
	Q3	120	0.958
	Q4	140	1.125

Forecast $\begin{cases} L+T+S \\ (L+T) \times S \end{cases}$